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THE COOPERMAN–PETRUNINA CHALLENGE SERIES

Five Open Problems Arising from the Algebraic Formulation of $\mathcal{Q}$-Numbers

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About this series

These five white papers each pose an open problem that grows out of “The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon” (Cooperman and Petrunina V.V., October 2026). Each is written for a different mathematical community, and each is stated precisely enough that nobody knows its answer and a proof or disproof would settle it.

No.	Problem	Audience
1	ℚ-Structured Yang–Mills Existence and Mass Gap	Mathematical physicists and gauge theorists
2	The ℚ-Structured Free Group Factor Problem	Operator algebraists and non-commutative geometers
3	The ℚ-Structured Simulability Dichotomy	Structural complexity theorists and algebraic geometers
4	The ℚ-Structured Spectral Gap Conjecture	Ergodic theorists and high-dimensional probabilists
5	The ℚ-Structured Exotic Detection Problem	Low-dimensional topologists and category theorists

ℚ-Structured Yang–Mills Existence and Mass Gap

Problem 1 of the Cooperman–Petrunina Challenge Series: for Mathematical Physicists and Gauge Theorists

Abstract

This white paper poses a well-defined problem of Millennium-Prize difficulty: does a ℚ-structured qubit regularization of lattice SU(2) gauge theory admit a continuum limit that is a quantum Yang–Mills theory with a strictly positive mass gap? The ℚ-structure, introduced by Cooperman and Petrunina V.V. (October 2026), imposes an antiunitary symmetry J_n whose square alternates with the parity of the qubit count n. A proof of the conjecture implies the Clay Millennium Problem for SU(2). A disproof would establish that ℚ-structured regularizations cannot reach quantum Yang–Mills theory. Its truth is unknown.

Lead-in

The Cooperman–Petrunina team is considering a challenge of Millennium-Prize difficulty. Before deciding, we want to set out exactly what it is, what it contains, and why we think the question is well-posed.

The starting point is a structural observation in "The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon" (Cooperman and Petrunina V.V., October 2026). The paper works with the real algebra ℚ spanned by {1, i, q, iq}, which is isomorphic to Hamilton's quaternions. It equips an n-qubit register with the antiunitary operator $J_n = (\sigma_y)^{\wedge}\otimes n\ K$, which satisfies:

$$J_n^2 = (-1)^n I$$

This is the time-reversal operator of n spin-½ degrees of freedom, and the sign is Kramers' theorem. Its consequence for the paper is concrete. For circuits whose generators commute with J_n, the dynamical Lie algebra lies in $sp(2^{n-1})$ when n is odd and in $so(2^n)$ when n is even. Dyson's threefold way appears as an alternation in the algebraic structure available to the system at each size.

Any non-perturbative construction of quantum Yang–Mills theory must pass through a regularization, and quantum simulation is now forcing one particular regularization into practice. A truncated Hamiltonian lattice gauge theory is encoded into qubits and then driven toward the continuum. In such a construction, the encoding is a choice. If the encoding carries a ℚ-structure, the regularization inherits the sp/so alternation at every finite stage.

This raises a question nobody has asked: does that structure survive the continuum limit, and does it help, hinder, or not affect the existence of a gapped quantum Yang–Mills theory?

We want to be explicit about the scale of this. The challenge asks for a continuum limit satisfying the Osterwalder–Schrader axioms with a strictly positive mass gap. That is the Clay Millennium Problem for SU(2), approached through a specific regularization. A positive answer would solve it. The ℚ-structure provides no shortcut; it adds conditions.

What the formulation does offer is a second outcome that the standard problem lacks. Suppose $\mathbb{Q}$-structure is incompatible with a gapped continuum limit, for example because the parity alternation leaves an imprint that cannot be renormalized away. Then that is a theorem about which regularizations can reach Yang–Mills at all, and it would matter immediately for quantum simulation of non-Abelian gauge theory.

Two points had to be settled before the statement could be made precise. First, for odd n, Kramers degeneracy doubles every energy level of a J_n-symmetric Hamiltonian. The gap must therefore be measured from the ground multiplet, not from the ground state. Second, the continuum limit sends the lattice spacing to zero, which is a different parameter from the qubit count. The statement keeps them separate and requires the limit to exist independently of the parity of n. A limit that existed only along odd n would not be a limit.

Nobody knows whether the conjecture is true. That is the point of posing it.

We would particularly value input from three communities:

- **Constructive field theorists:** whether $\mathbb{Q}$-structure is compatible with reflection positivity.
- **Lattice gauge theorists:** whether J_n can be made consistent with Gauss's law in a Kogut–Susskind truncation.
- **Operator algebraists:** what becomes of J_n, which in infinite volume survives only as an antilinear automorphism, in the limiting theory.

The Challenge

Setting

Kogut–Susskind Hamiltonian lattice SU(2) gauge theory on a three-dimensional spatial lattice $\Lambda(L, a)$, with side L, spacing a and coupling g(a). Link Hilbert spaces are truncated at representation cutoff j_max and encoded into n = n(L, a, j_max) qubits.

The encoding is **$\mathbb{Q}$-structured** if the encoded Hamiltonian H_n and every Gauss-law generator G_x commute with J_n:

$$[H_n, J_n] = 0,\ [G_x, J_n] = 0 \text{ for every lattice site } x, \text{ with } J_n = (\sigma_y)^{\otimes n} K$$

The **Kramers-reduced gap** $\Delta_K(n)$ is the distance from the ground multiplet of H_n, restricted to the gauge-invariant subspace, to the next distinct eigenvalue.

Conjecture

There exists a $\mathbb{Q}$-structured family of encodings and a renormalization trajectory g(a) with the following property. As $j_{max} \to \infty$, $L \to \infty$ and $a \to 0$, the vacuum states and dynamics converge, independently of the parity of n, to a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$. The limiting theory satisfies the Osterwalder–Schrader axioms and has a mass gap $\Delta > 0$, measured from the ground multiplet.

Prove or disprove

A proof must establish the conjecture as stated. A disproof must establish one of two things:

1. No $\mathbb{Q}$-structured encoding admits such a limit, with the obstruction identified.
2. The limit depends on the parity of n.

Subsidiary questions

These may be answered independently of the conjecture.

1. Does J_n induce an antiunitary symmetry of the limiting theory? If so, is it real or quaternionic?
2. Is $\mathbb{Q}$-structure compatible with reflection positivity at finite lattice spacing?

Scope and outcomes

Every resolution of the challenge is a substantial result; none is a known answer.

Outcome	What it establishes
Proof of the conjecture	Existence of quantum Yang–Mills theory for SU(2) on $\mathbb{R}^4$ with mass gap $\Delta > 0$, which implies the Clay Millennium Problem for SU(2)
Disproof via (1)	$\mathbb{Q}$-structured qubit regularizations cannot reach quantum Yang–Mills theory; the obstruction is identified
Disproof via (2)	The parity of the qubit count survives renormalization, so $\mathbb{Q}$-structured regularizations do not define a single continuum theory
Subsidiary question 1	Whether quaternionic time-reversal structure is a symmetry of continuum Yang–Mills or an artefact of the encoding
Subsidiary question 2	Whether $\mathbb{Q}$-structured encodings are admissible inputs for constructive field theory

This white paper makes no claim about the truth of the conjecture.

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The ℚ-Structured Free Group Factor Problem

Problem 2 of the Cooperman–Petrunina Challenge Series: for Operator Algebraists and Non-Commutative Geometers

Abstract

This white paper poses an open problem whose answer nobody knows: classify the free group factors over ℝ, ℂ and ℍ up to isomorphism. The ℚ-structure of Cooperman and Petrunina V.V. (October 2026) is the n-fold real tensor power of the quaternions, and its parity alternation is the order-2 class of ℍ in the Brauer group of ℝ. In the hyperfinite setting that parity cancels in the limit. In the free setting it survives as a genuine question about real forms, sitting on top of the free group factor isomorphism problem. The complex case alone is that problem, open since Murray and von Neumann. Octonions are excluded by a known theorem, which the paper states and uses.

Lead-in

To colleagues in von Neumann algebras, subfactors and non-commutative geometry: the Cooperman–Petrunina team is considering a second challenge, and we want to show you exactly how it was reached.

The starting point is "The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon" (Cooperman and Petrunina V.V., October 2026). The paper works with ℚ ≅ ℍ and the antiunitary $J_n = (\sigma_y)^{\otimes n} K$ on n qubits. Its Proposition 2 states that a Pauli rotation $\exp(-i\theta P)$ commutes with J_n exactly when the Pauli string P has odd weight. The proof is one line: conjugation by σ_y K flips the sign of every non-identity Pauli.

Operator algebraists will see the structure underneath at once. The real algebra fixed by Ad J_n is the n-fold real tensor power of the quaternions:

$$\mathbb{H}^{\otimes n} \cong M_2^{n}(\mathbb{R}) \text{ for } n \text{ even}, \quad \mathbb{H}^{\otimes n} \cong M_2^{n-1}(\mathbb{H}) \text{ for } n \text{ odd}.$$

The alternation the paper reports, the orthogonal algebra *so* for even n and the symplectic algebra *sp* for odd n, is the statement that [ℍ] has order 2 in the Brauer group $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2$. That is a clean, standard explanation, and it tells us immediately what happens in the hyperfinite limit. Grouping factors in pairs gives $\mathbb{H} \otimes \mathbb{H} \cong M_4(\mathbb{R})$, so the infinite tensor power of ℍ, closed in its trace representation, is the real hyperfinite II_1 factor. The parity cancels. In the amenable world, ℚ-structure leaves no trace.

We first looked at pushing one step further up the Cayley–Dickson ladder to the octonions. That route is closed by a known theorem. By the Jordan–von Neumann–Wigner classification and its JBW-algebra extension, the only octonionic factor is the exceptional Jordan algebra $H_3(\mathbb{O})$, and every purely exceptional JBW-algebra is of type I. Octonions cannot appear in a II_1 or type III limit, and exceptional factors admit no tensor products with other systems. Proposing an octonionic infinite tensor product as an open problem would be proposing something already ruled out.

The genuinely open territory is non-amenable. Define the real free group factor $L_{\mathbb{R}}(F_k)$ as the weak closure of the real span of the left-regular unitaries of the free group F_k, and the quaternionic one as $L_{\mathbb{H}}(F_k) = \mathbb{H} \otimes_{\mathbb{R}} L_{\mathbb{R}}(F_k)$. Their complexifications are known:

- $L_{\mathbb{R}}(F_k)$ complexifies to $L(F_k)$.
- $L_{\mathbb{H}}(F_k)$ complexifies to $M_2(L(F_k))$, which by the Dykema–Rădulescu amplification formula is the interpolated factor $L(F_{(k+3)/4})$.

So $L_{\mathbb{H}}(F_5)$ and $L_{\mathbb{R}}(F_2)$ have the same complexification, $L(F_2)$. Whether they are isomorphic as real von Neumann algebras is a question about real forms that complexification cannot see. It is the free-probability descendant of the $\mathbb{Q}$-parity: in finite dimensions Brauer theory decides it instantly, in the hyperfinite limit it disappears, and in the free setting, to our knowledge, nobody knows.

We pose it inside the full classification problem, which includes the complex case. That case is the free group factor isomorphism problem: is $L(F_2) \cong L(F_3)$? It has been open since Murray and von Neumann, and Dykema and Rădulescu showed the interpolated factors are either all isomorphic or pairwise distinct, without deciding which. Any complete answer to our challenge answers it.

We would particularly value input from three communities:

- **Free probabilists:** whether real (GOE) and quaternionic (GSE) matrix models carry information that complex (GUE) models do not. These are Dyson's $\beta = 1$ and $\beta = 4$ ensembles, the same threefold way that produced J_n.
- **Subfactor theorists:** whether the involutive anti-automorphisms defining these real forms are distinguished by any known invariant.
- **Real operator algebraists:** whether a Dykema–Rădulescu dichotomy holds for real and quaternionic interpolated free group factors.

The Challenge

Definitions

For $k \geq 2$, let λ be the left-regular representation of the free group F_k on $\ell^2(F_k)$.

- **Complex free group factor:** $L(F_k)$, the weak closure of the complex span of $\{\lambda(g) : g \in F_k\}$.
- **Real free group factor:** $L_{\mathbb{R}}(F_k)$, the weak closure of the real span of the same unitaries. Its complexification is $L(F_k)$.
- **Quaternionic free group factor:** $L_{\mathbb{H}}(F_k) = \mathbb{H} \otimes_{\mathbb{R}} L_{\mathbb{R}}(F_k)$. Its complexification is $M_2(L(F_k)) \cong L(F_{(k+3)/4})$.

Write $L_{\mathbb{K}}(F_k)$ for $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$. Isomorphism means isomorphism of real von Neumann algebras.

Main problem

Determine, for all $k, m \geq 2$ and all $\mathbb{K}, \mathbb{L} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, whether $L_{\mathbb{K}}(F_k) \cong L_{\mathbb{L}}(F_m)$.

A complete answer must settle both of the following.

1. **The classical question.** Is $L(F_2) \cong L(F_3)$?
2. **The parity question.** Is $L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$? Both complexify to $L(F_2)$, so this cannot be decided by complexification alone.

Prove or disprove

Either direction is a solution. A proof that the factors are pairwise non-isomorphic, a proof that some or all coincide, or a mixed classification each count, provided it covers every case of the main problem.

Subsidiary questions

These may be answered independently of the main problem.

1. Does a Dykema–Rădulescu dichotomy hold for real and quaternionic interpolated free group factors: are they either all isomorphic or pairwise distinct?
2. Does free entropy dimension computed with GOE ($\beta = 1$) or GSE ($\beta = 4$) matricial microstates agree with Voiculescu's GUE ($\beta = 2$) definition for every finite generating set?
3. For each k, is the canonical real structure on L(F_k) unique up to conjugacy among real forms, or does L(F_k) admit non-conjugate real forms?

Scope and outcomes

The main problem contains the free group factor isomorphism problem, so no complete answer is known and none can be reached without settling that problem.

Outcome	What it establishes
$L(F_2) \ncong L(F_3)$	The interpolated free group factors are pairwise distinct, by the Dykema–Rădulescu dichotomy, resolving the classical problem
$L(F_2) \cong L(F_3)$	All interpolated free group factors coincide, resolving the classical problem in the other direction
$L_{\mathbb{H}}(F_5) \ncong L_{\mathbb{R}}(F_2)$	The $\mathbb{Q}$-parity survives in non-amenable limits as an invariant invisible to complexification
$L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$	The $\mathbb{Q}$-parity cancels in free limits as it does in hyperfinite ones
Subsidiary question 2 answered	Whether Dyson's threefold way carries isomorphism information in free probability

What this challenge does not claim: octonionic infinite tensor products are excluded by known theorems and are not part of it. The open-status claims for the parity question and the

subsidiary questions reflect the authors' reading of the literature and should be checked before publication.

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The ℚ-Structured Simulability Dichotomy

Problem 3 of the Cooperman–Petrunina Challenge Series: for Structural Complexity Theorists and Algebraic Geometers

Abstract

This white paper poses an open problem whose answer nobody knows: for quantum circuits built from odd-weight Pauli rotations, is classical simulability decided exactly by the dimension of the dynamical Lie algebra? These are the ℚ-structured circuits of Cooperman and Petrunina V.V. (October 2026). Proving the hard direction unconditionally would separate P from PSPACE, a problem in the same class of difficulty as P versus NP. The easy-looking direction is also open, because polynomial Lie-algebraic simulation is known for expectation values but not for sampling. A constructive disproof is possible without any complexity breakthrough, so every route to a solution is concrete.

Lead-in

To colleagues in structural complexity and algebraic geometry: the Cooperman–Petrunina team is considering a third challenge, about where the boundary between classically simulable and genuinely quantum computation lies.

The starting point is “The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon” (Cooperman and Petrunina V.V., October 2026). The paper studies circuits whose generators are odd-weight Pauli strings, exactly the rotations that commute with the antiunitary $J_n = (\sigma_y)^{\otimes n} K$. Two results bracket the landscape.

At one extreme, a generic entangling ℚ-ansatz has the largest dynamical Lie algebra the symmetry allows:

$$\dim g = (4^n - (-2)^n) / 2$$

This is the dimension of the symplectic algebra $\mathrm{sp}(2^{n-1})$ for odd n and of the orthogonal algebra $\mathrm{so}(2^n)$ for even n, so the barren plateau survives at essentially the Haar rate. The quaternionic restriction alone does not cure it.

At the other extreme, the paper’s ℚ-cluster ansatz restricts to odd-weight monomials whose Lie closure is a free-fermion algebra of the form $\mathrm{so}(2a) \oplus \mathrm{so}(2b)$ with $a + b$ of order n, so its dimension is $\Theta(n^2)$. Through the Jordan–Wigner transformation these circuits are matchgate circuits, which Valiant, Terhal–DiVincenzo, Knill and Jozsa–Miyake showed are classically simulable.

It is tempting to conjecture that free fermions are the only tractable case. That is false, and the reason points to the real question. The single-qubit generators $Z_1, \ldots, Z_n$ are odd-weight, generate an abelian algebra of dimension n, and are trivially simulable. More generally, the g-sim method of Goh, Larocca, Cerezo and collaborators simulates expectation values for any dynamical Lie algebra of polynomial dimension. The natural dividing line is therefore not free fermions but the size of the algebra.

That leads to a clean, precise statement: within $\mathbb{Q}$-structured circuits, classical sampling is efficient if and only if the dynamical Lie algebra has polynomial dimension. Nobody knows whether it is true.

Each direction is hard for its own reason.

- **Exponential algebra implies hardness.** To prove this unconditionally, one must show that some quantum circuit family cannot be sampled classically in polynomial time. If P = PSPACE, every output probability could be computed exactly and every circuit sampled. An unconditional proof would therefore show P ≠ PSPACE, a separation in the same class of difficulty as P versus NP.
- **Polynomial algebra implies efficiency.** g-sim handles expectation values of observables inside the algebra. Sampling the full output distribution is a different task, and nobody has shown that a small algebra suffices for it.

The statement can also fail outright. A family with exponential algebra that can nevertheless be sampled efficiently would disprove it, and such a counterexample needs no complexity-theoretic breakthrough. The challenge is precise enough to be refuted, and hard enough that proving it would rank among the major results of the field.

We would particularly value input from three communities:

- **Structural complexity theorists:** whether existing classifications of commuting Hamiltonians and reversible gates extend to odd-weight Pauli generators.
- **Lie theorists and algebraic geometers:** a complete classification of the polynomial-dimension algebras that odd-weight Pauli sets can generate.
- **Quantum algorithm designers:** candidate families with exponential algebras that might still be classically sampleable.

The Challenge

Definitions

- **Odd-weight generators.** For each n, a set G_n of Pauli strings on n qubits, each of odd weight, computable in time polynomial in n. Every $\exp(-i\theta P)$ with P in G_n commutes with J_n.
- **$\mathbb{Q}$-structured circuit family.** For each n, circuits of the form $U = \exp(-i\theta_1 P_1) \cdots \exp(-i\theta_m P_m)$, with each P_j in G_n, m at most polynomial in n, and arbitrary real angles θ_j. The input is $|0\ldots0\rangle$ and the output is a computational-basis measurement.
- **Dynamical Lie algebra.** g_n is the real Lie algebra generated by $\{iP : P \in G_n\}$. It is a subalgebra of $so(2^n)$ or $sp(2^{n-1})$, according to the parity of n.
- **Efficiently sampleable.** A classical randomized algorithm, given a circuit description and $\varepsilon > 0$, outputs samples within total-variation distance ε of the circuit's output distribution, in time polynomial in n, m and $1/\varepsilon$.

Conjecture

For every $\mathbb{Q}$-structured circuit family, the following are equivalent:

1. The family is efficiently sampleable for all choices of angles.
2. dim g_n is bounded by a polynomial in n.

Prove or disprove

A proof must establish both directions unconditionally. A disproof must exhibit one of two things:

1. A family with dim g_n superpolynomial that is nevertheless efficiently sampleable.
2. A family with dim g_n polynomial that is provably not efficiently sampleable.

Subsidiary questions

These may be answered independently of the conjecture.

1. Assuming the polynomial hierarchy does not collapse, classify all odd-weight generator sets by the complexity of sampling their circuits.
2. Does polynomial dimension of g_n imply efficient sampling, beyond the expectation-value simulation that g-sim provides?
3. Classify all polynomial-dimension Lie algebras that sets of odd-weight Pauli strings can generate.
4. For each class in question 3, determine the scaling of gradient variance, in line with the Lie-algebraic theory of barren plateaus.

Scope and outcomes

A proof of the conjecture, or a disproof of the second kind, requires an unconditional classical lower bound and would therefore prove P ≠ PSPACE. A disproof of the first kind requires only a construction.

Outcome	What it establishes
Proof of the conjecture	Lie-algebra dimension exactly decides classical simulability for $\mathbb{Q}$-structured circuits, and P ≠ PSPACE
Disproof via (1)	A large Lie algebra does not imply quantum advantage; a new classical simulation technique beyond Lie-algebraic methods
Disproof via (2)	A small Lie algebra does not imply simulability, and P ≠ PSPACE
Subsidiary question 1	A conditional complexity classification of odd-weight Pauli gate sets
Subsidiary question 3	A complete map of the trainable, polynomial-dimension $\mathbb{Q}$-architectures

This white paper makes no claim about the truth of the conjecture. The open-status claims for the subsidiary questions reflect the authors' reading of the literature and should be checked before publication.

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The ℚ-Structured Spectral Gap Conjecture

Problem 4 of the Cooperman–Petrunina Challenge Series: for Ergodic Theorists and High-Dimensional Probabilists

Abstract

This white paper poses an open problem whose answer nobody knows: does every finite set of ℚ-structured quantum gates that generates a dense subgroup of its symmetry group have a spectral gap? For one qubit the symmetry group is Sp(1) = SU(2), the unit quaternions, and the conjecture is exactly the spectral gap conjecture for SU(2). That conjecture has been proved for gates with algebraic entries but is open for arbitrary rotation angles. The ℚ-structure of Cooperman and Petrunina V.V. (October 2026) extends the question to the full tower of groups $Sp(2^{n-1})$ and $SO(2^n)$, and ties it to the speed at which quantum circuits concentrate measure.

Lead-in

To colleagues in ergodic theory, measure concentration and spectral geometry: the Cooperman–Petrunina team is considering a fourth challenge, and it begins where barren plateaus meet harmonic analysis on compact groups.

The barren plateau phenomenon is concentration of measure in practice. As a parameterized quantum circuit becomes expressive, its distribution over states approaches a Haar-random design, and gradient variance decays exponentially in the number of qubits. “The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon” (Cooperman and Petrunina V.V., October 2026) computes this for circuits whose gates commute with the antiunitary $J_n = (\sigma_y)^{\otimes n} K$. Its Theorem 2 gives the loss variance of a deep, generic ℚ-structured circuit:

$$Var\ \ell = 1 / (2^n - (-1)^n)$$

The sign $(-1)^n$ records which group the circuit fills: the symplectic group $Sp(2^{n-1})$ when n is odd and the orthogonal group $SO(2^n)$ when n is even. Values of this kind follow from the Weingarten calculus for orthogonal and symplectic groups developed by Collins, Śniady and Matsumoto. The correction is a relative factor of $1 + O(2^{-n})$, so concentration holds in full. The parity changes the limiting measure slightly; it does not stop the circuit from reaching it.

The formula describes the destination, not the journey. How fast a random circuit approaches Haar measure is governed by the spectral gap of its averaging operator, and that is where the genuinely open mathematics lies.

The smallest case makes this concrete. For one qubit, $J_1^2 = -1$, and the gates commuting with J_1 form Sp(1) = SU(2), the group of unit quaternions. Take a finite set of such gates that generates a dense subgroup. Does the random walk they define have a spectral gap? This is the spectral gap conjecture for SU(2), raised by Sarnak and by Gamburd, Jakobson and Sarnak. Bourgain and Gamburd proved it when the gates have algebraic entries, and Benoist and de Saxcé extended that result to compact simple Lie groups. For arbitrary rotation angles, the case that quantum hardware actually produces, nobody knows.

Quaternions have already served this subject well. Lubotzky, Phillips and Sarnak used arithmetic quaternion algebras to build gate sets on SU(2) with optimal spectral gaps, and these "golden gates" now underpin efficient single-qubit gate synthesis. The $\mathbb{Q}$-structure extends the setting from Sp(1) to the whole tower $\mathrm{Sp}(2^{n-1})$ and $\mathrm{SO}(2^n)$, with the parity of n deciding which group appears.

The challenge below asks for the spectral gap at every level of that tower, for every dense generating set, with no algebraicity assumption. Its first case is a famous open problem. Its higher cases are harder.

We would particularly value input from three communities:

- **Additive combinatorialists:** whether the Bourgain–Gamburd method can be pushed past algebraic entries, even for generic angles.
- **Arithmetic geometers:** whether golden-gate constructions exist for $\mathrm{Sp}(2^{n-1})$ and $\mathrm{SO}(2^n)$ beyond the first case.
- **Quantum information theorists:** how fast local random $\mathbb{Q}$-circuits become approximate designs for these groups.

The Challenge

Definitions

- **Symmetry group.** For $n \geq 1$, K_n is the identity component of the group of unitaries on $(\mathbb{C}^2)^{\otimes n}$ that commute with J_n. It is $\mathrm{Sp}(2^{n-1})$ when n is odd and $\mathrm{SO}(2^n)$ when n is even.
- **$\mathbb{Q}$-structured gate set.** A finite set S of elements of K_n, closed under inverses, such that the subgroup generated by S is dense in K_n. No assumption is made on the entries of the gates.
- **Averaging operator.** T_S acts on $L^2(K_n, \mathrm{Haar})$ by $(T_S f)(g) = (1/|S|) \sum f(sg)$, the sum running over s in S.
- **Spectral gap.** S has a spectral gap if the operator norm of T_S on the orthogonal complement of the constant functions is strictly less than 1.

Conjecture

For every $n \geq 1$, every $\mathbb{Q}$-structured gate set S has a spectral gap.

Prove or disprove

A proof must cover every n and every dense generating set, with no algebraicity or diophantine assumption on the entries. A disproof must exhibit, for some n, a dense generating set of K_n whose averaging operator has norm 1 on the complement of the constants.

Subsidiary questions

These may be answered independently of the conjecture.

1. Let S_n be the odd-weight Pauli rotations of weight at most 3 at a fixed algebraic angle. Whenever S_n generates a dense subgroup and $n \geq 3$, Benoist–de Saxcé gives it a spectral gap; determine how that gap scales with n, polynomially or exponentially.

2. Determine whether golden-gate sets with optimal, Ramanujan-type spectral gaps exist for Sp(4), the n = 3 case. (For n = 2, SO(4) is covered by two copies of SU(2), so the Lubotzky–Phillips–Sarnak construction applies.)
3. Show that brickwork circuits of local $\mathbb{Q}$-structured gates form ε-approximate t-designs for K_n in depth polynomial in n and t, the symmetric analogue of Brandão–Harrow–Horodecki.
4. Derive the full Weingarten expansion of the $\mathbb{Q}$-circuit loss variance, identifying every place the parity $(-1)^n$ enters.

Scope and outcomes

The n = 1 case of the conjecture is the spectral gap conjecture for SU(2), so any proof settles that problem, and any disproof found at n = 1 refutes it.

Outcome	What it establishes
Proof of the conjecture	Every dense $\mathbb{Q}$-structured gate set mixes at an exponential rate, at every n; this includes the spectral gap conjecture for SU(2)
Disproof at n = 1	The spectral gap conjecture for SU(2) is false: some dense set of unit quaternions mixes arbitrarily slowly
Disproof at some $n \geq 2$	Spectral gaps can fail in the higher $\mathbb{Q}$-structured groups even if they hold for SU(2)
Subsidiary question 2	Optimal gate synthesis for the n = 3 symmetry group Sp(4)
Subsidiary question 3	A provable depth bound for when $\mathbb{Q}$-circuits reach their barren-plateau regime

This white paper makes no claim about the truth of the conjecture, nor does it verify the paper's Theorem 2. The open-status claims for the subsidiary questions reflect the authors' reading of the literature and should be checked before publication.

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The ℚ-Structured Exotic Detection Problem

Problem 5 of the Cooperman–Petrunina Challenge Series: for Low-Dimensional Topologists and Category Theorists

Abstract

This white paper poses an open problem whose answer nobody knows: can a 4-dimensional topological quantum field theory, built from finite categorical data carrying the ℚ-structure of Cooperman and Petrunina V.V. (October 2026), tell apart two closed smooth 4-manifolds that are homeomorphic but not diffeomorphic? No TQFT built from finite algebraic data has ever done this. Reutter proved that semisimple ones cannot, so any solution must work outside semisimplicity. A ℚ-structured invariant that took a non-standard value on a homotopy 4-sphere would disprove the smooth 4-dimensional Poincaré conjecture, the last open case of the Poincaré conjecture.

Lead-in

To colleagues in low-dimensional topology and higher category theory: the Cooperman–Petrunina team is considering a fifth challenge, aimed at the hardest frontier in the field, smooth 4-manifolds.

Quantum invariants transformed the topology of knots and 3-manifolds. The Jones polynomial and the Witten–Reshetikhin–Turaev invariants come from braided tensor categories, and they distinguish objects that classical tools cannot. In dimension 4 the story stalls. Distinguishing homeomorphic but non-diffeomorphic smooth 4-manifolds requires gauge theory and its descendants: Donaldson, Seiberg–Witten and Heegaard Floer invariants. None of these comes from a TQFT built out of finite algebraic data.

This is not for lack of trying, and two theorems explain why. Freedman, Kitaev, Nayak, Slingerland, Walker and Wang showed that the universal manifold pairing fails to be positive in dimension 4, so unitary 4-dimensional TQFTs cannot distinguish certain smooth structures. Reutter then proved that semisimple 4-dimensional TQFTs cannot detect exotic smooth structure on simply connected closed 4-manifolds at all. Any finite-data invariant that sees exotic structure must therefore be non-semisimple, and if it is to escape the positivity obstruction, non-unitary.

The ℚ-numbers paper enters here. “The Algebraic Formulation of ℚ-Numbers in Quantum Information Systems and Mitigation of the Barren Plateau Phenomenon” (Cooperman and Petrunina V.V., October 2026) studies the antiunitary $J_n = (\sigma_y)^{\otimes n} K$ on n qubits, which satisfies $J_n^2 = (-1)^n$. Read physically, this is a time-reversal symmetry with $T^2 = (-1)^F$, where F counts the qubits. Kapustin, Thorngren, Turzillo and Wang showed that topological phases with this symmetry are described by field theories on Pin manifolds, which may be non-orientable. Mathematically, the alternation is the Brauer-group fact that quaternionic tensor quaternionic is real.

We should be precise about what this structure alone provides. Representations with a real or quaternionic structure form a symmetric tensor category, and the TQFTs it produces directly are semisimple, so Reutter’s theorem applies to them. ℚ-structure by itself cannot detect exotic

smoothness. What it offers is a symmetry type, and with it access to non-orientable manifolds, which oriented theories never see. It must be combined with genuinely non-semisimple categorical input.

Non-orientable 4-manifolds are a natural place to look. Cappell and Shaneson constructed a 4-manifold homotopy equivalent to $\mathbb{R}P^4$ but not diffeomorphic to it, and $\mathbb{R}P^4$ carries the Pin structures the $\mathbb{Q}$-symmetry calls for. Homotopy 4-spheres built by Cappell–Shaneson and Gluck constructions are the highest-stakes test cases. If any invariant distinguished one of them from the standard 4-sphere, the smooth 4-dimensional Poincaré conjecture would be false.

The challenge below asks whether a non-semisimple, $\mathbb{Q}$-structured 4-dimensional TQFT can detect exotic smooth structure. Nobody knows. A construction would be the first finite-data invariant of exotic smoothness. A proof that none exists would extend Reutter's theorem past semisimplicity, which is itself open.

We would particularly value input from three communities:

- **Higher category theorists:** non-semisimple finite tensor categories and 2-categories carrying a Pin-type time-reversal structure.
- **Smooth 4-manifold topologists:** exotic pairs, non-orientable or simply connected, whose difference might be visible to a Pin-structured invariant.
- **Quantum topologists:** whether recent non-semisimple constructions in dimension 3 can be lifted to dimension 4 with a $\mathbb{Q}$-structure.

The Challenge

Definitions

- **Finite 4-dimensional TQFT.** A symmetric monoidal functor from a bordism category of 4-manifolds to finite-dimensional vector spaces, defined by finite categorical data. It need not be semisimple or unitary.
- **$\mathbb{Q}$-structured TQFT.** A finite 4-dimensional TQFT defined on Pin manifolds, whose state spaces carry an antilinear involution T with $T^2 = (-1)^F$, the categorical form of $J_n^2 = (-1)^{\underline{n}}$.
- **Exotic pair.** Two closed smooth 4-manifolds X and X′ that are homeomorphic but not diffeomorphic.
- **Detection.** A TQFT Z detects an exotic pair if $Z(X) \neq Z(X')$ for some choice of compatible Pin structures.

Main problem

Determine whether there exists a $\mathbb{Q}$-structured TQFT that detects some exotic pair.

Prove or disprove

Either answer is a solution.

1. **Existence.** Construct a $\mathbb{Q}$-structured TQFT and an exotic pair, and prove that the TQFT distinguishes them.

2. **Non-existence.** Prove that no $\mathbb{Q}$-structured TQFT detects any exotic pair. This requires extending Reutter's theorem beyond semisimple theories.

Subsidiary questions

These may be answered independently of the main problem.

1. Does any $\mathbb{Q}$-structured TQFT take a value on a Cappell–Shaneson or Gluck-twist homotopy 4-sphere that differs from its value on the standard 4-sphere? A positive answer disproves the smooth 4-dimensional Poincaré conjecture.
2. Does any $\mathbb{Q}$-structured TQFT distinguish Cappell–Shaneson's homotopy $\mathbb{RP}^4$ from $\mathbb{RP}^4$?
3. Does Reutter's theorem hold for semisimple TQFTs on Pin manifolds, so that semisimple $\mathbb{Q}$-structured theories are also blind to exotic structure?

Scope and outcomes

No TQFT built from finite algebraic data is known to detect exotic smooth structure, and nobody knows whether one can. Every outcome below would be a first.

Outcome	What it establishes
Existence proved	The first finite-data invariant of exotic smoothness in dimension 4, and a new tool alongside gauge theory
Non-existence proved	Finite TQFTs with this symmetry are blind to exotic structure, extending Reutter's theorem past semisimplicity
Subsidiary question 1 answered positively	The smooth 4-dimensional Poincaré conjecture is false
Subsidiary question 2 answered positively	Pin-structured invariants see smooth structure that oriented invariants cannot reach
Subsidiary question 3 answered	The semisimple $\mathbb{Q}$-structured case is settled, in either direction

This white paper makes no claim about the answer. The open-status claims for the subsidiary questions reflect the authors' reading of the literature and should be checked before publication.

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